DATA ANALYSIS STRATEGY & PLANNING

Project FOODFLOW

Decoding production efficiency in virtual food processing with advanced Excel

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Primary tool: Microsoft Excel (Power Query, PivotTables, dynamic arrays, what-if tools)

THE PLAN AT A GLANCE

33 hrs estimated workload across 8 phases	**5** public data sources, cross-checked	**6** research questions, 4 testable hypotheses	**5** data-quality gates, borrowed from HACCP

The big idea

A food factory is a production line: raw material comes in, gets inspected, processed, packed and dispatched. **This project runs the analysis itself like that line**, with five "stations" that turn messy public data into clean, trusted, decision-ready insight. Quality gates borrowed from food safety (HACCP) protect the data the same way they protect the product.

1. Project Introduction

1.1 What is "virtual food processing"?

Food processing turns raw agricultural inputs (grain, milk, oilseeds, fruit, meat) into safe, shelf-ready products. **Virtual food processing** means studying and optimising that chain through data and simulation rather than on a physical shop floor: a "digital twin lite" of a processing plant, built from public statistics and tested in a spreadsheet sandbox before any real money or machinery is touched.

For this project, I model a fictional mid-sized plant, **NutriLine Foods**, whose inputs, throughput, labour, energy use and margins are benchmarked against publicly reported industry data. The plant is illustrative, but the numbers it is compared with are real.

1.2 Why data-driven decisions matter here

- **Thin margins.** Food manufacturing earns modest margins, so small efficiency gains (less waste, less energy per tonne) matter a great deal.
- **Volatile inputs.** Crop yields, weather and commodity prices swing, and processors must plan procurement and production around that volatility.
- **Perishability and safety.** Spoilage, yield loss and compliance failures are expensive. Data helps spot problems before they become recalls or write-offs.
- **Rising pressure on energy and sustainability.** Energy intensity and waste are now both a cost line and a reporting requirement.

1.3 Why Excel?

Excel is the most widely used analysis tool in operations and finance teams. Modern Excel (Power Query, dynamic arrays, Power Pivot, what-if tools) can clean, model, forecast and visualise a project of this size end to end, and every step stays transparent and auditable, which is exactly what a planning exercise needs.

1.4 The numbers behind the case

The plan is anchored in published figures. These are the real data points that motivated the research questions, quoted as reported by each source.

Indicator	Published figure	Source and why it matters
Value added per worker, US food manufacturing (2019)	Food (NAICS 311) made up **7.6%** of US manufacturing employment but **9.1%** of value added. That is roughly **1.2x** the manufacturing average value added per worker (9.1 / 7.6).	US Census Bureau, Annual Survey of Manufactures, via StatsAmerica. Shows that efficiency can be measured and compared across segments (RQ2).
US food manufacturing employment	**1,477.8** thousand jobs (2010) to **1,554.1** thousand (2015): **+5.2%** in five years, about **1.0%** a year.	US Bureau of Labor Statistics industry productivity series (via FRED). Real input for the growth-rate indicator in Section 6.2.
India: food products in registered factories (2023-24)	Food products employed **11.06%** of factory workers, about **21.6 lakh** people, and ranks among the top five industries by gross value added.	MoSPI Annual Survey of Industries 2023-24, as reported in press coverage. Why an Indian lens belongs in the plan.

Indicator	Published figure	Source and why it matters
Revisions to official data	FAO's 2025 world cereal output forecast moved from **2,911** million tonnes (June 2025 brief) to **3,023** million tonnes in a later brief: a **+3.8%** revision.	FAO Cereal Supply and Demand Brief. Why every download is date-stamped (CCP4) and headline figures are re-checked before reporting.
Excel capacity	**1,048,576** rows per worksheet.	Microsoft Excel specifications. Why large source files are filtered in Power Query before loading.

These figures will be re-checked against the primary tables during Phase 2, because published statistics are revised over time.

2. Objectives, Research Questions and Hypotheses

2.1 Analysis objectives

ID	Objective (SMART)
O1	**Consolidate** at least five public datasets into one cleaned, documented Excel data model covering a minimum of 10 years where data allows.
O2	**Quantify trends** in production volume and processing output across five food categories (cereals, dairy, edible oils, fruit and vegetables, meat), including seasonality and volatility.
O3	**Build a Production Efficiency Index (PEI)**: a single composite score combining labour productivity, energy intensity and yield, so segments and years can be ranked fairly.
O4	**Test four hypotheses** about what drives efficiency, using correlation, regression and lag analysis inside Excel.
O5	**Deliver an interactive dashboard and at least five evidence-backed recommendations** for the NutriLine Foods virtual plant, each with a what-if value estimate.

2.2 Key research questions

ID	Research question
RQ1	How have production volumes of key food categories moved over the last 10 to 15 years, and which categories are the most seasonal or volatile?
RQ2	Which food-manufacturing segments create the most value added per worker and per unit of energy?
RQ3	Do raw-material price changes squeeze processing margins, and if so, after what delay?
RQ4	Does higher capital intensity (investment per worker) translate into higher productivity?
RQ5	Where is the largest efficiency gap between NutriLine Foods and industry benchmarks, and what would closing it be worth?
RQ6	What does the next three years of output look like under base, optimistic and stressed scenarios?

2.3 Working hypotheses

ID	Hypothesis	How Excel will test it
H1	Capital-intensive segments produce more output per worker.	Scatter plot with trendline, CORREL, and LINEST regression (capital per worker vs output per worker).
H2	Raw-material price spikes reduce processing margins with a lag of one to two periods.	Lagged CORREL on offset ranges; conditional formatting to highlight months where price rose and margin fell.
H3	Segments with higher energy use per tonne score lower on overall efficiency.	PEI ranking table, Data Analysis ToolPak regression, and a colour-scaled heat map.
H4	Staple categories (cereals, oils) are more stable than perishable ones (fruit and vegetables, dairy).	Standard deviation of year-on-year % change per category (STDEV.S), compared in a PivotChart. CV kept as a secondary check.

2.4 Decision rules, set in advance

Each hypothesis gets a pass or fail threshold before any data is analysed, so the verdict cannot be bent to fit the result.

ID	Supported if…
H1	Correlation (r) between capital per worker and output per worker is **0.5 or higher**, and the LINEST slope is significant at the 5% level ($p < 0.05$).
H2	The lag-1 or lag-2 correlation between raw-material price change and margin change is **-0.5 or stronger**. With 15 annual observations, $\|r\|$ must exceed about **0.51** to be significant at the 5% level.
H3	Energy per tonne correlates with efficiency at **r of -0.5 or stronger**. To avoid circular logic (energy is inside the PEI), this is tested against the PEI recalculated **without** the energy component.
H4	The **median country ratio** of volatility (perishable divided by staple) is **at least 2x**, where volatility is the standard deviation of year-on-year % change. CV is reported only as a secondary check, because long-run growth inflates it (see Section 7.1).

Verdicts: **supported** if the rule is met, **partly supported** if it holds for some segments, regions or lags but not all, and **not supported** otherwise.

3. Scope and Limitations

In scope	Out of scope
Publicly available annual and monthly data on food production, prices, and food-manufacturing performance. Three lenses: India, the United States and the European Union, for comparison. Five food categories, plus an "all food manufacturing" aggregate. A fictional virtual plant (NutriLine Foods) used as the benchmark target. All analysis, modelling and dashboards built in Excel.	Proprietary or plant-level operational data. Real-time sensor (IoT) data and live production feeds. Machine-learning or predictive models outside Excel. Food-safety laboratory results and regulatory compliance audits. Detailed financial modelling of real companies.

Limitations to be honest about

- **Aggregated, not plant-level.** Public data describes industries and countries, so NutriLine Foods is a calibrated illustration and not a prediction of any real plant.
- **Definitions differ across sources.** Industry classifications, units and currencies vary, so harmonisation (see data-quality gates) is necessary and introduces assumptions.
- **Reporting lags.** Official manufacturing surveys are often published with a delay of a year or more, so the latest period may be missing for some sources.
- **Excel scale limits.** A worksheet holds just over one million rows. Large source files will be filtered at import (Power Query) rather than loaded whole.
- **Correlation is not causation.** Findings will be framed as evidence and hypotheses to investigate, not proof.

4. Data Source Identification

Five public sources were chosen so that each research question has at least two independent sources behind it. Together they cover **what is produced** (volumes), **what it costs** (prices), and **how well factories convert inputs into outputs** (efficiency).

Source	What it provides	Why it was selected	Used for
FAOSTAT (UN FAO) fao.org/faostat	Production quantities, yields and producer prices by country and commodity, with long annual history.	The global standard for food production data. Free, consistent across countries, and downloadable as CSV.	RQ1, RQ3, RQ6 and H2, H4
US Census Bureau: Annual Survey of Manufacturers census.gov/programs-surveys/asm	Food manufacturing (NAICS 311): value of shipments, value added, employment, payroll, capital expenditure, energy and materials cost.	The richest public view of how factories convert inputs to outputs. It supplies the labour, capital and energy measures behind the PEI.	RQ2, RQ4, RQ5 and H1, H3
India: Annual Survey of Industries (MoSPI) and data.gov.in data.gov.in	Factory-level aggregates for food product manufacturing: output, workers, fixed capital, fuel and materials.	Brings an Indian lens, relevant to a fast-growing processing sector, and allows a developing-economy versus developed-economy comparison.	RQ2, RQ4, RQ5
Eurostat (Structural Business Statistics and PRODCOM) ec.europa.eu/eurostat	Food, beverage manufacturing turnover, employment, production value and volumes by EU country.	Provides a third regional benchmark with harmonised definitions across many countries, which strengthens cross-checks.	RQ2, RQ5, RQ6
Open Food Facts world.openfoodfacts.org	Crowd-sourced product database: categories, ingredients and nutrition for packaged food worldwide.	Adds a product-mix layer to the story, so segments can be profiled by product type and processing level.	Context and segment profiling (RQ2)

Selection logic

Official first. Government and UN sources lead, because their methods are documented and their data can be cited.

Triangulate. Efficiency measures from the US, India and EU are compared, so no single country's definitions drive the conclusions.

Free and Excel-friendly. Every source offers CSV or XLSX download, so the full pipeline stays reproducible.

Check before use. Table names, release dates and download formats change over time, so exact tables and licence terms will be confirmed and logged in the workbook during Phase 2.

5. Methodology: The FOODFLOW Line

The analysis moves through five stations, just like a product moving down a processing line. Each station has a clear output, and nothing moves forward until it passes its gate.

Station	Food-line analogy	What happens in the analysis
1. Intake	Receiving raw material	Download datasets, record source, date and licence, and load everything through Power Query into one workbook.
2. Inspect	Quality control checks	Run the five data-quality gates (below). Reject, repair or flag anything that fails.
3. Process	Cooking and transforming	Standardise units and currency, build the master table, calculate ratios and the Production Efficiency Index.
4. Package	Packing and labelling	Create PivotTables, charts, heat maps and an interactive dashboard with slicers.
5. Dispatch	Shipping to customers	Turn findings into recommendations with what-if values and a short executive summary.

5.1 Data-quality gates (inspired by HACCP)

HACCP identifies "critical control points" where food safety must be checked. The same idea applies to data: a few well-placed checks catch most errors before they reach a chart.

Gate	Question asked	Excel check	If it fails
CCP1	Is the data complete?	=COUNTBLANK(range)/ROWS(range) must be under 5% per key column.	Fill using a documented method, or exclude the series.
CCP2	Is it consistent?	Units converted to tonnes and currencies converted to USD in Power Query. Category names mapped through a lookup table.	Fix the mapping table and refresh.
CCP3	Is it plausible?	Z-score formula flags values beyond three standard deviations. Conditional formatting highlights them in red.	Verify against the source. Keep, correct or annotate.
CCP4	Is it traceable?	Every row carries Source and Download_Date columns. A Log sheet records every change.	Block the data from the master table until sourced.

Gate	Question asked	Excel check	If it fails
CCP5	Is it repeatable?	All cleaning is done as Power Query steps, never by hand edits.	Rebuild the step in Power Query.

5.2 Step-by-step method

1. **Frame.** Confirm objectives, research questions and hypotheses. Agree what "efficient" means (the PEI components and weights).
2. **Collect.** Download the five datasets, record their source, date and licence in a README sheet.
3. **Load and clean.** Import with Power Query. Remove duplicates, unpivot wide year columns, standardise units, currencies and category names.
4. **Inspect.** Run the five data-quality gates and record results in the Log sheet.
5. **Build the master table.** Merge sources into one Excel Table keyed by Year, Region and Segment.
6. **Engineer indicators.** Calculate output per worker, value added per worker, energy per tonne, material cost share, yield, growth rates and volatility.
7. **Compute the PEI.** Normalise each component to a 0 to 1 scale, apply weights held in named assumption cells, and rank segments and years.
8. **Explore and test.** Build PivotTables for trends and seasonality, then test H1 to H4 with CORREL, LINEST and the Data Analysis ToolPak.
9. **Forecast and stress-test.** Use FORECAST.ETS for three-year outlooks, and Data Tables and Scenario Manager for base, optimistic and stressed cases.
10. **Optimise (what-if).** Use Goal Seek and Solver to estimate the value of closing NutriLine's efficiency gaps.
11. **Visualise.** Assemble the dashboard: KPI cards, slicers, trend lines, heat map and gap-closure waterfall.
12. **Conclude.** Write findings, recommendations and limitations. Peer-check formulas and numbers.

6. The Advanced Excel Toolkit

Tool or technique	Where it is used	Why it earns its place
Power Query	Import, unpivot, merge and clean all five sources.	Repeatable steps, so refreshing with new data takes one click.
Excel Tables and structured references	Master data table and every calculation sheet.	Formulas read like sentences and expand automatically with new rows.
XLOOKUP / INDEX-MATCH	Category mapping, benchmark lookups.	Two-way, error-tolerant lookups with no column-order fragility.
SUMIFS / AVERAGEIFS / COUNTIFS	Ratios by region, segment and year.	Multi-criteria aggregation without pivot dependence.
Dynamic arrays (FILTER, SORT, UNIQUE, LET)	Live leaderboards and top-N segment lists.	Results spill and update automatically, with no copy-down formulas.
PivotTables, PivotCharts, Slicers and Timelines	Trends, seasonality and segment comparison.	Fast exploration, and the dashboard becomes interactive.
Conditional formatting	Outlier flags, heat maps, icon-set KPIs, data bars.	Patterns and problems become visible at a glance.

Tool or technique	Where it is used	Why it earns its place
Data validation	Assumption cells and dashboard selectors.	Prevents invalid inputs from breaking the model.
Statistical functions and ToolPak	CORREL, STDEV.S, LINEST, regression output.	Tests the hypotheses with real statistics.
FORECAST.ETS and trendlines	Three-year outlooks with seasonality.	Time-series forecasting without leaving Excel.
Data Tables, Scenario Manager, Goal Seek, Solver	Base, optimistic and stressed scenarios, and gap-closure value.	Turns analysis into decisions: "what if we cut energy use by 10%?"
Sparklines and charts	Compact trend lines in KPI cards and scatter plots with trendlines.	Dense, readable communication of results.

6.1 Signature formulas

Examples of the formulas that sit at the heart of the model (cell references are illustrative):

```
Normalised score (0 to 1):   =(C2-MIN(C$2:C$40))/(MAX(C$2:C$40)-MIN(C$2:C$40))
PEI (weights = named cells):  =w_labour*N2 + w_energy*O2 + w_yield*P2
YoY change (per row):       =IF(A3=A2,(D3-D2)/D2,"")
Volatility (YoY SD):        =STDEV.S(E3:E57)
Volatility (CV, secondary): =STDEV.S(D2:D16)/AVERAGE(D2:D16)
Lagged correlation (1 per): =CORREL(D3:D40, E2:E39)
Outlier flag (CCP3):        =ABS((C2-AVERAGE(C$2:C$40))/STDEV.S(C$2:C$40))>3
Live top-5 leaderboard:     =TAKE(SORT(FILTER(Master,Master[Year]=sel_yr),7,-
1),5)
Three-year forecast:        =FORECAST.ETS(A41,B2:B40,A2:A40,1)
```

The PEI weights (for example 40% labour productivity, 30% energy efficiency, 30% yield) are deliberately held in labelled input cells, so the sensitivity of the ranking to those choices can itself be tested.

6.2 Worked examples

> **How to read these examples**
>
> **Example A uses real published figures** (BLS employment). **Examples B to D use illustrative numbers** to demonstrate the method on a six-row mini dataset. In the real project, these inputs are replaced by the cleaned data from Section 4, and the formulas stay the same.

Example A: growth rate from real data. US food manufacturing employment was 1,477.8 thousand in 2010 and 1,554.1 thousand in 2015 (BLS). The compound annual growth rate is:

```
=(1554.1/1477.8)^(1/5)-1     // = 1.01%, about 1.0% a year (total change +5.2%)
```

Example B: building the Production Efficiency Index. Weights: 40% labour productivity, 30% energy efficiency, 30% yield. Energy is "lower is better", so it is inverted when normalised.

Segment	Output (t/labour hr)	Energy (kWh/t)	Yield (%)	Norm. output	Norm. energy	Norm. yield	PEI	Rank
Edible oils	6.5	140	88	1.000	0.686	0.739	**0.827**	1
Cereal milling	5.1	85	76	0.696	1.000	0.217	**0.643**	2
Dairy	3.2	210	94	0.283	0.286	1.000	**0.499**	3
NutriLine Foods	3.0	230	80	0.239	0.171	0.391	**0.264**	**4**
Meat	2.4	190	71	0.109	0.400	0.000	**0.163**	5
Fruit and veg	1.9	260	82	0.000	0.000	0.478	**0.143**	6

Working for NutriLine Foods:

```
Norm. output = (3.0-1.9)/(6.5-1.9)   = 0.239
Norm. energy = (260-230)/(260-85)    = 0.171   // inverted: lower is better
Norm. yield  = (80-71)/(94-71)       = 0.391
PEI = 0.4*0.239 + 0.3*0.171 + 0.3*0.391 = 0.264
```

Reading the result: NutriLine ranks 4th of 6. Its weakest component is energy (0.171 against a best of 1.000), which suggests starting with energy. Section 8.2 then checks whether that is also where the money is.

Example C: what-if value of a 10% energy cut. Energy falls from 230 to 207 kWh/t, so the normalised energy score becomes (260-207)/175 = 0.303 and the PEI rises from 0.264 to **0.304** (+0.040, a 15% improvement). The rank stays 4th. Assuming (for illustration) 20,000 tonnes a year and electricity at $0.10 per kWh:

```
Annual saving = 23 kWh/t x 20,000 t x $0.10 = $46,000 a year
```

Goal Seek answers the reverse question: what energy use would lift NutriLine to Dairy's PEI of 0.499? The answer is about **93 kWh/t, a 59% cut**, which is unrealistic. So energy alone cannot close the gap, and yield and labour productivity must improve too. That is the kind of finding the what-if panel is designed to surface.

Example D: data-quality gates in action.

- **CCP1 (completeness):** a key column has 4 blanks in 60 rows. =COUNTBLANK(D2:D61)/ROWS(D2:D61) gives 6.7%, above the 5% limit, so the column **fails** and must be repaired or excluded.

- **CCP3 (plausibility):** a column has mean 100 and standard deviation 12. A value of 142 has z = (142-100)/12 = **3.5**, above 3, so it is flagged red and checked against the source before it is kept or corrected.

Testing the hypotheses. The full step-by-step tests for H1 to H4, including a pilot on real FAO data, are in Section 7. Robustness checks and scenarios are in Section 8.

6.3 Dashboard design

- **Top row:** four KPI cards (average PEI, output per worker, energy per tonne, forecast growth) with icon-set arrows and sparklines.
- **Controls:** slicers for Year, Region and Segment, plus a timeline for monthly data.
- **Centre:** trend line for production volume, colour-scaled heat map of PEI by segment and year.
- **Bottom:** scatter of capital per worker against output per worker (H1) and a waterfall chart showing the value of closing each NutriLine gap.

7. Testing the Hypotheses: Worked Step by Step

Every hypothesis gets the same treatment: the data, the Excel steps with formulas, the result, the pre-set decision rule from Section 2.4, and a verdict.

> **What is real and what is illustrative**
>
> **H4 is run on real FAO data** (142 countries, 1961 to 2013), so its numbers are genuine findings from a pilot.
>
> **H1 to H3 need factory-level, price and energy data** that are collected in Phase 2. Until then they use small illustrative datasets to show the exact steps and formulas. In the real run the cleaned master table replaces these inputs and nothing else changes.

7.1 H4: are staples steadier than perishables? (real FAO data)

Data: FAO diet compositions (kilocalories per person per day) from Our World in Data, 1961 to 2013, using the 142 countries with complete series. Staple = cereals and grains. Perishables = fruit and vegetables, dairy and eggs, and meat.

1. **Arrange the data** as Country, Year, Category, kcal, sorted by Country, Category, then Year.
2. **Year-on-year change:** =IF(A3=A2,(D3-D2)/D2,"") so the calculation restarts at each new country and category.
3. **Volatility:** the standard deviation of those changes, built as a PivotTable (Values field set to StdDev) by Country and Category.
4. **Ratio per country:** perishable volatility divided by cereal volatility. Summarise with =MEDIAN(ratio) and =COUNTIF(ratio,">=2")/COUNT(ratio).

Worked example with real numbers: India, 2008 to 2013 (kcal per person per day).

Year	Cereals	YoY change	Fruit and vegetables	YoY change
2008	1,418	-	113	-
2009	1,381	-2.61%	110	-2.65%
2010	1,404	+1.67%	119	+8.18%
2011	1,386	-1.28%	122	+2.52%
2012	1,344	-3.03%	124	+1.64%
2013	1,361	+1.26%	131	+5.65%
STDEV.S		**2.17%**		**4.12%**

For this six-year window the ratio is 4.12 / 2.17 = **1.90**. Over India's full 1961 to 2013 series, though, cereals vary by 5.06%, fruit and vegetables by 4.14% and dairy and eggs by 4.91%, so the ratio falls to **0.82**. The answer depends on the window, which is why the test uses the full period and many countries.

Result across 142 countries:

Category	Median volatility (YoY SD)	Median ratio to cereals	Countries with ratio > 1	Countries with ratio >= 2
Cereals and grains (staple)	4.72%	-	-	-
Fruit and vegetables	8.15%	1.85	82.4%	42.3%
Dairy and eggs	8.56%	1.96	85.2%	48.6%
Meat	7.10%	1.34	76.8%	31.0%

Why not the coefficient of variation? CV is inflated by long-run growth. India's fruit and vegetable supply more than doubled (55 to 131 kcal), giving a CV of 26.9% even though its year-on-year volatility is only 4.1%. Across countries the median CV ratios (about 2.0 to 2.1) would have looked like an easy pass. The pilot therefore moved the primary metric to year-on-year volatility, and this change is logged as a method refinement.

Verdict: PARTLY SUPPORTED. The direction holds: perishables are more volatile than cereals in 77% to 85% of countries. The size falls short of the pre-set 2x rule (median ratios 1.3 to 2.0), and India is a counter-example over the full period. One caution: this is food supply per person, which trade and stocks smooth for cereals, not production. Phase 2 repeats the test on FAOSTAT production quantities.

7.2 H1: do capital-intensive segments produce more per worker? (illustrative data)

Observation	1	2	3	4	5	6	7	8	9	10	11	12
Capital per worker ($k)	40	55	62	75	80	95	110	120	135	150	165	180
Output per worker ($k)	95	120	105	140	150	135	170	160	210	190	235	225

1. **Set up:** capital per worker in column B, output per worker in column C, one row per segment-year (12 here).
2. **Look first:** Insert a scatter chart, then add a trendline showing the equation and R-squared.
3. **Correlation:** =CORREL(B2:B13,C2:C13) gives **r = 0.957**.
4. **Line of best fit:** =SLOPE(C2:C13,B2:B13) = **0.972**, =INTERCEPT(...) = **58.65**, =RSQ(...) = **0.915**. Reading: each extra $1,000 of capital per worker goes with about $972 more output per worker, and capital explains 91.5% of the variation.
5. **Significance:** t = r * SQRT(n-2) / SQRT(1-r^2) = 0.957 * SQRT(10) / SQRT(1-0.915) = **10.41**. Then =T.DIST.2T(10.41,10) gives **p = 0.000001**. With n = 12, |r| only needs to exceed 0.576 to be significant at 5%.
6. **Decision rule:** r >= 0.5 and p < 0.05. Both are met.

Verdict: SUPPORTED (on illustrative data). The real test also needs a check that large segments do not drive the whole pattern, and a reminder that correlation is not causation.

7.3 H2: do raw-material price spikes squeeze margins with a lag? (illustrative data)

Year (20xx)	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Price change (%)	4	-2	12	8	-5	3	15	-8	2	6	-3	10	5	-6	9
Margin change (pp)	0.4	0.0	-0.9	-2.8	-4.2	2.9	-1.8	-2.9	3.1	-2.0	-0.7	-0.8	-2.1	0.0	0.8

1. **Set up:** price change in B2:B16, margin change in C2:C16, one row per year (2010 to 2024).

2. **Same-year link (lag 0):** =CORREL(B2:B16,C2:C16) = **0.081**, essentially none.
3. **Margin one year later (lag 1):** =CORREL(B2:B15,C3:C16) = **-0.795** on 14 pairs. Shifting the ranges by one row is the whole trick.
4. **Two years later (lag 2):** =CORREL(B2:B14,C4:C16) = **0.470** on 13 pairs.
5. **Critical values:** lagging loses pairs, so the bar rises a little: |r| must exceed **0.532** for 14 pairs and **0.553** for 13. Only lag 1 clears it (p = 0.0007).
6. **Size of the effect:** =SLOPE(C3:C16,B2:B15) = **-0.24**. A 10% rise in raw-material price goes with a margin about 2.4 points lower the next year.
7. **Make it visible:** a conditional formatting rule =AND(B2>10,C3<-2) shades the years where price jumped by more than 10% and the next year's margin fell by more than 2 points. It flags 2012 and 2016.

Verdict: SUPPORTED at a one-year lag (r = -0.795, needed -0.5 or stronger). Lag 0 and lag 2 do not pass, which fits the hypothesis of a delayed effect.

7.4 H3: do energy-hungry segments score lower on efficiency? (illustrative data)

Energy is already inside the PEI, so correlating energy with the full index would be circular. The test uses the index recalculated **without** energy, with the labour and yield weights scaled to sum to 1: PEI ex-energy = (0.4 x norm output + 0.3 x norm yield) / 0.7.

Segment	Energy (kWh/t)	PEI without energy	Rank
Edible oils	140	0.888	1
Dairy	210	0.590	2
Cereal milling	85	0.491	3
NutriLine Foods	230	0.304	4
Fruit and veg	260	0.205	5
Meat	190	0.062	6

1. **Recalculate the index without energy** as above (this uses the normalised values from Example B).
2. **Correlation:** =CORREL(energy, PEI ex-energy) = **-0.497**.
3. **Sample-size check:** with only six segments, |r| must exceed **0.811** to be significant at 5%. The p-value is **0.316**.
4. **Decision rule:** r of -0.5 or stronger. -0.497 just misses, and it is not significant anyway.
5. **Fix for the real run:** use a segment-by-year panel. Six segments over ten years gives 60 observations, where |r| above about **0.25** is enough.

Verdict: NOT SUPPORTED on six points. The direction is negative as predicted, but six observations cannot separate a moderate relationship from chance. For comparison, using the full PEI gives r = -0.762, which looks impressive and is exactly why it must not be used: energy is inside it.

8. Original Analysis: Robustness and Scenarios

A ranking is only useful if it survives questions. This section stress-tests the NutriLine Foods result from Example B and turns it into decisions. All inputs here are illustrative and sit in the Assumptions sheet.

8.1 Does the ranking depend on the weights?

Same data as Example B, five different weight sets (labour / energy / yield). In Excel this is a one-variable Data Table with the weight set as the input.

Weights	NutriLine PEI	Rank	Leader (PEI)	Gap to Dairy
Base 40 / 30 / 30	0.264	4	Edible oils (0.827)	0.235
Equal 33 / 33 / 33	0.267	4	Edible oils (0.808)	0.256
Energy-heavy 20 / 50 / 30	0.251	4	Edible oils (0.765)	0.248
Labour-heavy 60 / 20 / 20	0.256	4	Edible oils (0.885)	0.171
Yield-heavy 20 / 20 / 60	0.317	4	Edible oils (0.781)	0.397

Reading: NutriLine ranks 4th under all five weight sets, so the finding is robust. What the weights change is **how far** it trails Dairy: the gap runs from 0.17 to 0.40, and it is widest when yield is weighted heavily because Dairy has the best yield. The weights decide the size of the gap, not whether it exists.

8.2 Which lever pays most?

Assumptions (illustrative): 20,000 tonnes of input a year, electricity at $0.10 per kWh, and a selling price of $900 per tonne of output.

Lever	Change	PEI after	Change in PEI	Annual value
Cut energy use	230 to 207 kWh/t (-10%)	0.304	+0.040	+$46,000 energy cost saved
Raise yield	80% to 83%	0.304	+0.040	+$540,000 (600 t more output)
Raise labour productivity	3.0 to 3.3 t per labour hour (+10%)	0.291	+0.027	Not monetised in this model
All three together		**0.369**	**+0.105**	**+$586,000**

Reading: energy is NutriLine's weakest component, but **yield is the better first target**. It lifts the PEI by the same 0.040 and is worth about 12 times as much money ($540,000 against $46,000). A ranking tells you where you are weakest. The what-if tells you where fixing it pays. Both belong in the recommendation.

8.3 Base, optimistic and stressed scenarios

Scenario	Output (t/hr)	Energy (kWh/t)	Yield (%)	PEI	Rank	Net annual value
Stressed	2.85	253	78	0.186	4	-$406,000 (energy +$46,000, yield -400 t)
Base	3.0	230	80	**0.264**	4	$0
Optimistic	3.3	207	83	0.369	4	+$586,000

Reading: the downside (-$406,000) and upside (+$586,000) are lopsided in favour of improvement, which supports acting. Even the optimistic case keeps NutriLine 4th, because Dairy (0.499) is out of reach through

these three levers alone. The stressed case still ranks above Meat (0.163) and Fruit and veg (0.143). In Excel, Scenario Manager stores the three cases and produces this table in one click.

9. Timeline and Resource Planning

The plan totals **33 hours**, inside the 30 to 35 hour estimate. It is built for **10 working sessions of about three hours each**, which can run over two to three weeks. Two hours are held back as a buffer in Phase 8 and the larger phases (P3, P5) have slack inside them.

9.1 Task breakdown

Phase	Key tasks	Hours	Deliverable
P1. Frame and plan	Confirm objectives, questions, hypotheses, PEI definition. Set up workbook structure and README.	2	Approved scope and workbook skeleton
P2. Intake: find and collect data	Locate exact tables, download five datasets, record source, date and licence, and run a first look at each file.	5	Raw data sheets and source log
P3. Inspect and clean	Power Query cleaning, unit and currency standardisation, category mapping, and CCP1 to CCP5 checks.	5	Clean master table and quality log
P4. Process: model and indicators	Engineer ratios, normalise, build the PEI, add NutriLine benchmark profile.	4	Calculation sheets and PEI ranking
P5. Analyse and test	PivotTables, seasonality and volatility, correlation and regression for H1 to H4, forecasts and scenarios.	7	Findings sheet and hypothesis verdicts
P6. Package: dashboard	Build charts, heat maps, slicers, KPI cards and what-if panel. Test the interactions.	4	Interactive dashboard
P7. Dispatch: insights and report	Write insights, recommendations with what-if values, limitations and executive summary.	4	Final report and recommendations
P8. Quality assurance and buffer	Formula audit, number cross-check, peer feedback, polish and contingency.	2	Verified, submission-ready workbook

9.2 Timeline view (D = working session)

Phase	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Hours
Frame and plan											2
Intake: find and collect data											5
Inspect and clean											5
Process: model and indicators											4
Analyse and test											7
Package: dashboard											4
Dispatch: insights and report											4
Quality assurance and buffer											2
Total											**33**

9.3 Resources

Resource	Detail
Software	Microsoft Excel (365 recommended for dynamic arrays, Power Query and Power Pivot). Enable the Data Analysis ToolPak and Solver add-ins.
Data	Five free public sources listed in Section 4. All are downloadable as CSV or XLSX.
People	One analyst (me), plus a peer or mentor for a review in Phase 8.
Documentation	README and Log sheets inside the workbook, with this plan as the governing document.
Cost	No data or licensing cost.

10. Risk Assessment

Each risk is scored for likelihood (L) and impact (I) on a 1 to 3 scale. Priority = L x I, so anything scoring 6 or more gets an active mitigation from day one.

Risk	L	I	Priority	Mitigation
Inconsistent definitions across sources	3	3	**9**	Build the category mapping table early, document every assumption, and compare at index level (growth and ratios) as well as at absolute level.
Missing or delayed data	3	2	**6**	Set a 5% completeness threshold (CCP1), use documented interpolation only for short gaps, and exclude series that fail rather than guess.
Source data revised after download	3	2	**6**	Date-stamp every download (CCP4) and re-pull headline series before the final report. FAO's 2025 cereal forecast moved +3.8% between briefs, so revisions are normal and should be expected.

Risk	L	I	Priority	Mitigation
Spreadsheet formula errors	2	3	**6**	Use Tables and named ranges, separate inputs from calculations, add check cells, and have a peer audit in Phase 8.
Large files slow Excel	2	2	4	Filter and aggregate in Power Query before loading, and use the Data Model rather than raw sheets for big tables.
Currency and inflation distortions	2	3	**6**	Convert to a single currency, use real (inflation-adjusted) values for trend analysis, and note the base year.
Over-reading correlations	2	3	**6**	State findings as evidence, not proof. Check the lags, test with more than one source and show the r-squared.
Scope creep	2	2	4	Hold to the five questions and five sources. Park extra ideas in a "Phase 2 ideas" list.
Time overrun	2	2	4	Use the 2-hour buffer, timebox each phase, and cut dashboard extras before cutting quality checks.
Data licence or access changes	1	2	2	Keep a dated copy of each download in the workbook and note the licence terms in the README.

11. Expected Outcomes and Success Criteria

11.1 What the finished project will deliver

- A single, documented Excel workbook that anyone can refresh with new data.
- A Production Efficiency Index that ranks segments and years transparently.
- A clear verdict (supported, partly supported, not supported) on each of the four hypotheses.
- An interactive dashboard and five or more recommendations, each with a what-if value.

11.2 How success will be judged

Criterion	Test
Comprehensive	Every research question is answered with data from at least two sources, and every objective has a visible output.
Relevant	Every tab and technique ties back to food processing efficiency and a named question.
Clear	A reader can follow the logic from question to method to result without a guide, and the dashboard can be understood in under two minutes.
Original	HACCP-style data gates, a custom efficiency index, a fictional virtual plant benchmarked against real data, and what-if valuation of every recommendation.
Reproducible	Refreshing Power Query with new data updates every table and chart without manual repair.

11.3 Measurable targets

Target	Measure
Data coverage	At least 5 sources loaded and at least 10 years of history where available.

Target	Measure
Data quality	Every key column under 5% blanks (CCP1) or documented as excluded; every outlier above |z| = 3 reviewed and logged.
Transparency	Zero hard-coded numbers in the PEI sheet: all weights and assumptions sit in named input cells.
Analysis	Four hypotheses each given a verdict against a pre-set rule; six research questions each answered with a chart or table.
Recommendations	At least five, each with a what-if value (for example, an annual saving or a PEI change).
Effort	Total between 30 and 35 hours, with the 2-hour buffer unspent or documented.

Closing thought

Good analysis planning is a recipe, not a guess: know what you are cooking (objectives), what goes into it (data), how you will prepare it (method), how long it takes (timeline), and what could burn (risks). This plan puts all five on the table before the first formula is typed.

Appendix: Workbook Blueprint

The planned structure of the Excel file, so the build stays organised from day one.

Sheet	Purpose
README	Project summary, source list with dates and licences, sheet guide, and version notes.
Assumptions	PEI weights, base year, currency rates and scenario inputs, all as named cells with data validation.
Raw_FAO / Raw_ASM / Raw_ASI / Raw_EU / Raw_OFF	Untouched downloads, one sheet per source. Never edited by hand.
Map	Lookup tables that map categories, countries and units across sources.
Master	Clean, merged Excel Table keyed by Year, Region and Segment.
Calc_PEI	Indicators, normalisation and the Production Efficiency Index.
Tests	Correlation, regression, lag analysis, and hypothesis verdicts.
Scenarios	Forecasts, Data Tables, Goal Seek and Solver what-if models.
Pivots	PivotTables and PivotCharts feeding the dashboard.
Dashboard	KPI cards, slicers, charts and heat map, presented for the reader.
Log	Data-quality gate results and a record of every data decision.